

# SURYAVARDAN SURESH

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## Education

**New York University (Graduate School of Arts and Science)**

**Aug 2023 - May 2025**

*MS in Computer Science (Courant)*

*GPA: 4.00*

**Indian Institute of Information Technology (IIIT) Sri City**

**Aug 2019 - May 2023**

*B.Tech in CSE with Honors (Transfer Learning and GANs)*

*GPA: 8.96*

## Skills

**Technical Skills:** Python (Pytorch, Tensorflow, Numpy, Matplotlib, Django etc.), R (Shiny), Matlab, Java (MERN), SQL, C/C++, HTML/CSS/JS/Bootstrap, Github

**Coursework:** [MS] Conceptual Gaps in Modern ML, Cloud and ML, Predictive Analytics, OS, Fundamental Algorithms, [BTech] ML, NLP, OOPS, DSA, Web Development, Adv. Engineering Mathematics, Communication Skills, DBMS etc.

**Certifications:** Deep Learning Specialisation, Build Better GANs, Natural Language Processing, Web Development for Everybody, Machine Learning with Python

## Experience

**Intuitive Surgical**

**May 2024 - Present**

*Data Scientist Intern*

*R, Shiny, Forecasting*

- Supporting the manufacturing team by leveraging data analytics, forecasting, data engineering and data science to solve problems and optimize supply chain, logistics, operations and planning processes.
- Delivered solutions and applications that helped in maintaining continuity of supply, reducing line-downs and created a dashboard to track manufacturing KPIs. Includes Barcode scanner based apps, R dashboards etc.

**Artificial Intelligence Institute of UofSC**

**Aug 2022 - Present**

*Research Intern*

*Deep Learning, Python, Tensorflow*

- Orchestrated the development of Joint Embeddings to out-perform industry benchmarks in Image Captioning, Text-based-Image retrieval, and Image2Tweet tasks, delivering a 6% boost in BLEU score and culminating in the following paper: **IMAGINATOR: Pre-Trained Image+Text Joint Embeddings using Word-Level Grounding of Images**

**Wipro AI Lab**

**Apr 2021 - Aug 2022**

*Research Intern*

*Data Science, Deep Learning, Python, PyTorch*

- Led the Organizing Committee for the De-Factify workshop 🤖 at AAI, overseeing all aspects from task formulation to data collection, cleaning, organization, annotation, analysis, baseline model creation, participants' paper review, and conference presentations.
- Delivered a compelling presentation on the Factify shared task, detailing the meticulous curation and classification process of 100,000 tweets and news articles across 5 classes based on relative entailment. More about it in **(FACTIFY: A Multi-Modal Fact Verification Dataset @ AAI 2022)**

## Projects

**Honors Research - B.Tech** | *Tensorflow, Matlab, Research*

- Introduced a novel GAN variation to analyze the effects of synthetic data on traditional Unsupervised Domain Adaptation algorithms; delivered enhancements on 62% of tasks outlined within current SOTA benchmarks. This work was presented at the ICONIP-22 conference, publication: **(UDA Supplemented with Generated Images - ICONIP 2022)**

**Deep Learning** | *Pytorch, Langchain, Tensorflow, HPC*

- Pioneered a QnA-chatbot on LENR-research data and compared performance of SOTA open-source LLMs with grid search on parameters, under the supervision of Prof. Anasse Bari and Prof. David Nagel [ongoing paper]
- Earned top 15 percentile in the Deep Learning bootcamp 🤖 conducted by DPhi; Kaggle, VarDial etc.

**Metro Simulation** 🤖 | *Agent-based modelling, Netlogo*

- Formulated an advanced ABM tool to simulate and evaluate metro architecture of any city, featuring excel input for routes and passenger count; Washington DC metro data study conducted to uncover optimization strategies for congestion and wait times, enhancing urban transportation planning.

**Web Development** | *React, Node.js, MongoDB, REST API, Docker, Django, SQL, JS, Bootstrap*

- Spearheaded the creation of ShoeBox 🤖, an e-commerce platform for shoes infused with social media elements, seamlessly integrating APIs for enhanced pagination and filtering capabilities, implementing Redux for streamlined state management, and documenting the process.
- Engineered and executed a comprehensive Donation Platform 🤖 for the sem project, encompassing Admin, Donor, and NGO modules.